CAMBRIDGE IGCSE™ & O LEVEL COMPUTER SCIENCE (0478 / 2210)

STUDENT MASTER STUDY GUIDE

TOPICS 1.1.4, 1.1.5 & 1.1.6 (Binary Arithmetic, Logical Shifts & Two's Complement)

TOPIC 1.1.4: ADDITION OF BINARY NUMBERS & OVERFLOW ERROR

1. The 4 Fundamental Rules of Binary Addition

Operation	Result	Explanation / Action
0 + 0	**0**	Zero with no carry
0 + 1 (or 1 + 0)	**1**	One with no carry
1 + 1	**10 (in binary)**	Write **0** in current column, **Carry 1** to the next left column
1 + 1 + 1	**11 (in binary)**	Write **1** in current column, **Carry 1** to the next left column

2. Worked Example (Step-by-Step)

Task: Add 01011010 (90 in denary) and 00110111 (55 in denary):

```
Carry bits:   1 1 1 1 1 0 0
              0 1 0 1 1 0 1 0   (Denary: 90)
            + 0 0 1 1 0 1 1 1   (Denary: 55)
            -----------------
     Result:  1 0 0 1 0 0 0 1   (Denary: 145)
```

• **Verification:** 90 + 55 = 145. (128 + 16 + 1 = 145). Result is correct!

3. Understanding Overflow Error

• An 8-bit register can only store numbers from **0 to 255**.
• An **Overflow Error** occurs when the sum of two binary numbers exceeds 255. A 9th carry bit is generated from the Most Significant Bit (MSB).
• Because an 8-bit register has no space to store this 9th bit, the bit is dropped (truncated), causing an incorrect and inaccurate calculation result.

TOPIC 1.1.5: LOGICAL BINARY SHIFTS (LEFT & RIGHT)

1. Logical Left Shift (× 2 for each place shifted)

• **Action:** All bits move to the left. The vacated spaces on the right are always filled with **0**.
• **Mathematical Effect:** Shifting left by 1 place **multiplies the number by 2**. Shifting left by n places multiplies by 2^n.
• **Example:** Original value = `00110100` (Denary: 52).
Shift 2 places left → `11010000` (Denary: 128 + 64 + 16 = 208 = 52 × 4).
• **Overflow Risk:** If a '1' bit is shifted out of the Most Significant Bit (MSB), an **overflow error** occurs, causing loss of data accuracy.

2. Logical Right Shift (÷ 2 for each place shifted)

• **Action:** All bits move to the right. The vacated spaces on the left are always filled with **0**.
• **Mathematical Effect:** Shifting right by 1 place **divides the number by 2** (integer division).
• **Example:** Original value = `00011000` (Denary: 24).
Shift 1 place right → `00001100` (Denary: 12).
• **Precision Risk:** If an odd number is shifted right (e.g. 13 ÷ 2 = 6.5), the '1' bit in the LSB is shifted out and lost, causing **loss of precision** (fractional part is lost).

TOPIC 1.1.6: TWO'S COMPLEMENT (SIGNED NEGATIVE NUMBERS)

1. 8-Bit Signed Place Value Chart (Range: -128 to +127)

-128 (MSB)	64	32	16	8	4	2	1 (LSB)
Sign Bit (1 = Negative, 0 = Positive)	Magnitude Bits (Values 64 down to 1)						

2. How to Convert Negative Denary to Two's Complement Binary

Example: Convert -45 into 8-bit Two's Complement Binary:
• **Step 1:** Write +45 in 8-bit binary → `00101101`
• **Step 2:** Invert all bits (One's Complement: change 0 to 1 and 1 to 0) → `11010010`
• **Step 3:** Add 1 to the result → `11010010 + 1 = 11010011` (Final Answer: **11010011**)

Quick Exam Shortcut (Right-to-Left Rule):
1. Take positive binary: `0 0 1 0 1 1 0 1`
2. Starting from the right (LSB), copy all bits up to and including the first '1': `... 1`
3. Invert all remaining bits to the left: `1 1 0 1 0 0 1 1`

3. How to Convert Two's Complement Binary to Denary

Example: Convert 11001000 into Denary:
• Use place values: (-128 × 1) + (64 × 1) + (32 × 0) + (16 × 0) + (8 × 1) + (4 × 0) + (2 × 0) + (1 × 0)
• Calculation: -128 + 64 + 8 = **-56**.